

## Creating Microservices for account and loan

### Accounts Microservice:

#### Accounts.java

```
package com.cognizant.account.model;

public class Account {

    private String number;

    private String type;

    private double balance;

    public Account() {

    }

    public Account(String number, String type, double balance) {

        this.number = number;

        this.type = type;

        this.balance = balance;

    }

    public String getNumber() {

        return number;

    }

    public void setNumber(String number) {

        this.number = number;

    }

    public String getType() {

        return type;

    }

    public void setType(String type) {
```

```

        this.type = type;
    }

    public double getBalance() {

        return balance;
    }

    public void setBalance(double balance) {

        this.balance = balance;
    }
}

```

### **AccountController.java**

```

package com.cognizant.account.controller;

import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;
import com.cognizant.account.model.Account;

@RestController

public class AccountController {

    @GetMapping("/accounts/{number}")

    public Account getAccount(@PathVariable String number) {

        return new Account(

            number,

            "savings",

            234343

        );
    }
}

```

```
}
```

### **AccountApplication.java**

```
package com.cognizant.account;

import org.springframework.boot.SpringApplication;

import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication

public class AccountApplication {

    public static void main(String[] args) {

        SpringApplication.run(AccountApplication.class, args);

    }

}
```

### **Pom.xml**

```
<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-
4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <parent>

        <groupId>org.springframework.boot</groupId>

        <artifactId>spring-boot-starter-parent</artifactId>

        <version>4.1.0</version>

        <relativePath/> <!-- lookup parent from repository -->

    </parent>

    <groupId>com.cognizant</groupId>

    <artifactId>account</artifactId>
```

```
<version>0.0.1-SNAPSHOT</version>

<name/>

<description/>

<url/>

<licenses>

  <license/>

</licenses>

<developers>

  <developer/>

</developers>

<scm>

  <connection/>

  <developerConnection/>

  <tag/>

  <url/>

</scm>

<properties>

  <java.version>21</java.version>

</properties>

<dependencies>

  <dependency>

    <groupId>org.springframework.boot</groupId>

    <artifactId>spring-boot-starter-webmvc</artifactId>

  </dependency>

  <dependency>
```

```
        <groupId>org.springframework.boot</groupId>

        <artifactId>spring-boot-devtools</artifactId>

        <scope>runtime</scope>

        <optional>true</optional>
    </dependency>

    <dependency>

        <groupId>org.springframework.boot</groupId>

        <artifactId>spring-boot-starter-webmvc-test</artifactId>

        <scope>test</scope>
    </dependency>
</dependencies>

<build>

    <plugins>

        <plugin>

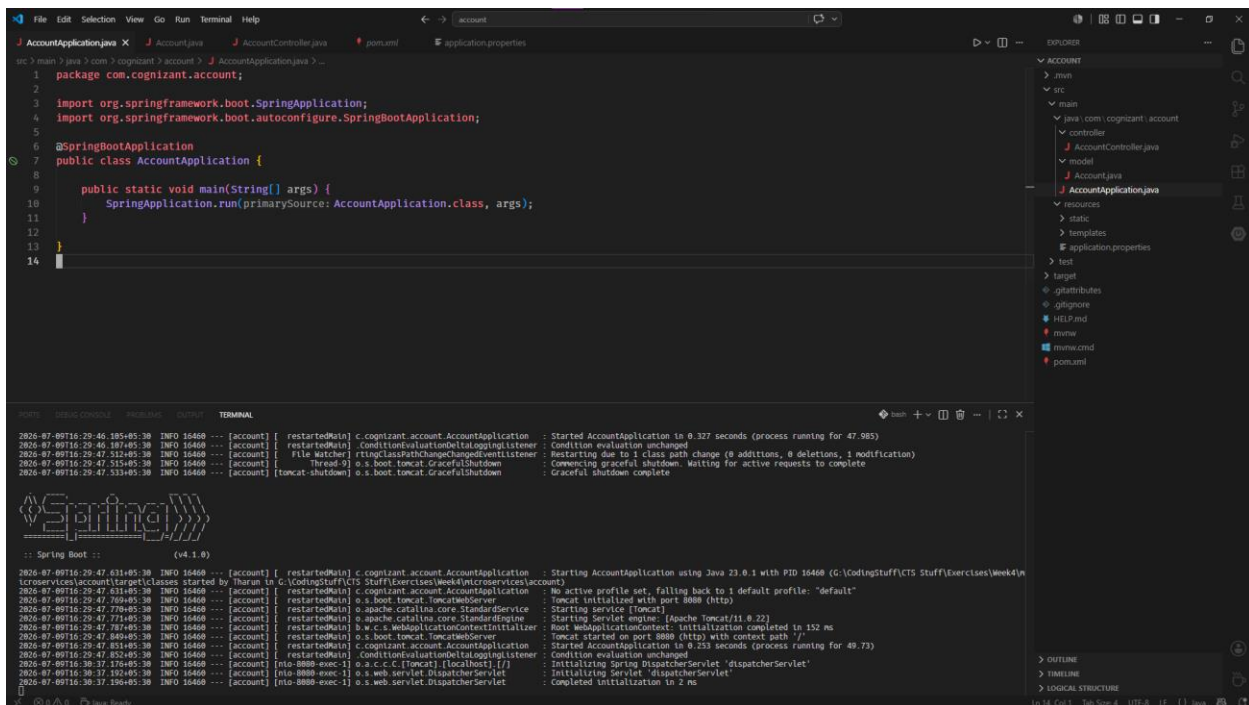
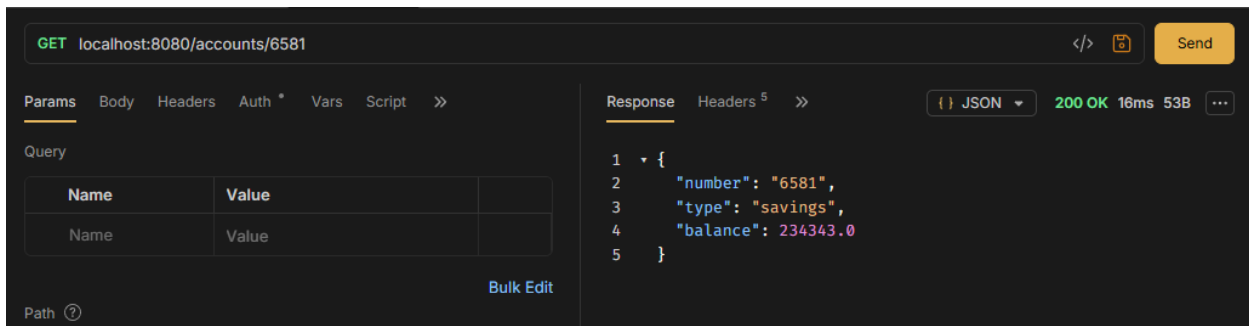
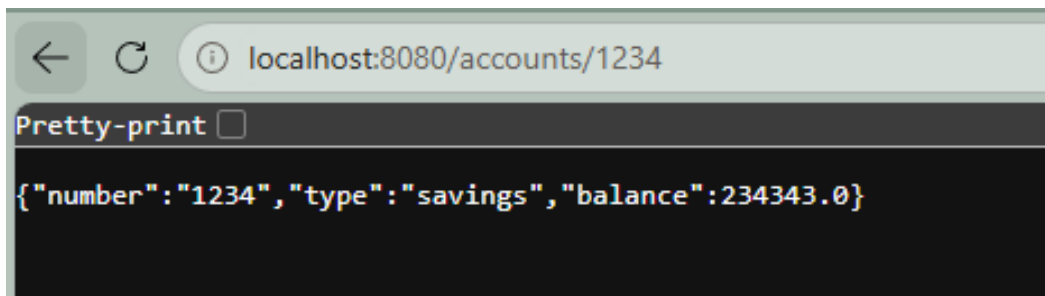
            <groupId>org.springframework.boot</groupId>

            <artifactId>spring-boot-maven-plugin</artifactId>
        </plugin>
    </plugins>
</build>
</project>
```

### **Application.properties**

spring.application.name=account

server.port=8080



## Loan Microservice

Loan.java

```
package com.cognizant.loan.model;
```

```
public class Loan {
```

```
private String number;
```

```
private String type;
```

```
private double loan;
```

```
private double emi;
```

```
private int tenure;
```

```
public Loan() {
```

```
}
```

```
public Loan(String number, String type,
```

```
    double loan,
```

```
    double emi,
```

```
    int tenure) {
```

```
    this.number = number;
```

```
    this.type = type;
```

```
    this.loan = loan;
```

```
    this.emi = emi;
```

```
    this.tenure = tenure;
```

```
}
```

```
public String getNumber() {
```

```
    return number;
```

```
}
```

```
public void setNumber(String number) {
```

```
    this.number = number;
```

```
}
```

```
public String getType() {  
    return type;  
}  
  
public void setType(String type) {  
    this.type = type;  
}  
  
public double getLoan() {  
    return loan;  
}  
  
public void setLoan(double loan) {  
    this.loan = loan;  
}  
  
public double getEmi() {  
    return emi;  
}  
  
public void setEmi(double emi) {  
    this.emi = emi;  
}  
  
public int getTenure() {  
    return tenure;  
}  
  
public void setTenure(int tenure) {  
    this.tenure = tenure;  
}  
}
```



### **LoanController.java**

```
package com.cognizant.loan.controller;

import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;
import com.cognizant.loan.model.Loan;

@RestController

public class LoanController {

    @GetMapping("/loans/{number}")

    public Loan getLoan(@PathVariable String number) {

        return new Loan(

            number,

            "car",

            400000,

            3258,

            18);

    }

}
```

### **LoanApplication.java**

```
package com.cognizant.loan;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication

public class LoanApplication {

    public static void main(String[] args) {
```

```
        SpringApplication.run(LoanApplication.class, args);
    }
}
```

### **Pom.xml**

```
<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-
4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <parent>

    <groupId>org.springframework.boot</groupId>

    <artifactId>spring-boot-starter-parent</artifactId>

    <version>4.1.0</version>

    <relativePath/> <!-- lookup parent from repository -->

  </parent>

  <groupId>com.cognizant</groupId>

  <artifactId>loan</artifactId>

  <version>0.0.1-SNAPSHOT</version>

  <name/>

  <description/>

  <url/>

  <licenses>

    <license/>

  </licenses>

  <developers>
```

```
<developer/>
</developers>
<scm>
  <connection/>
  <developerConnection/>
  <tag/>
  <url/>
</scm>
<properties>
  <java.version>21</java.version>
</properties>
<dependencies>
  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-webmvc</artifactId>
  </dependency>

  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-devtools</artifactId>
    <scope>runtime</scope>
    <optional>true</optional>
  </dependency>
  <dependency>
    <groupId>org.springframework.boot</groupId>
```

```
<artifactId>spring-boot-starter-webmvc-test</artifactId>

<scope>test</scope>

</dependency>

</dependencies>

<build>

  <plugins>

    <plugin>

      <groupId>org.springframework.boot</groupId>

      <artifactId>spring-boot-maven-plugin</artifactId>

    </plugin>

  </plugins>

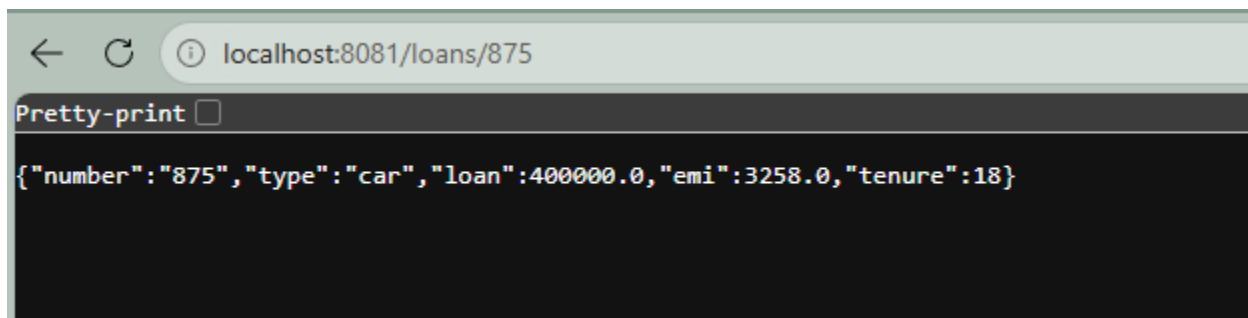
</build>

</project>
```

### Application.properties

spring.application.name=loan

server.port:8081



GETlocalhost:8081/loans/8124

</>Send

ParamsBodyHeadersAuthVarsScript

Query

| Name | Value |
|------|-------|
| Name | Value |

Bulk Edit

ResponseHeaders

JSON200 OK7ms71B

```
1 {
2   "number": "8124",
3   "type": "car",
4   "loan": 400000.0,
5   "emi": 3258.0,
6   "tenure": 18
7 }
```

FileEditSelectionViewGoRunTerminalHelp

Loan.javaLoanController.javaapplication.propertiesLoanApplication.java

src > main > java > com > cognizant > loan > LoanApplication.java

```
1 package com.cognizant.loan;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6 @SpringBootApplication
7 public class LoanApplication {
8
9     public static void main(String[] args) {
10         SpringApplication.run(LoanApplication.class, args);
11     }
12 }
13
14
```

Run [Debug]

2026-07-09T16:32:32.529+05:30 INFO 14688 ... [loan] [ restartedMain] com.cognizant.loan.LoanApplication : Starting LoanApplication using Java 23.0.1 with PID 14688 (G:\Coding\Stuff\CTIS\Stuff\Exercises\Week4\Microservices\Loan)
2026-07-09T16:32:32.533+05:30 INFO 14688 ... [loan] [ restartedMain] com.cognizant.loan.LoanApplication : No active profile set, falling back to 1 default profile: 'default'
2026-07-09T16:32:32.634+05:30 INFO 14688 ... [loan] [ restartedMain] o.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! Set 'spring.devtools.add-pr
ng\_level.web' property to 'false' to disable
2026-07-09T16:32:32.634+05:30 INFO 14688 ... [loan] [ restartedMain] o.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'loggi
2026-07-09T16:32:33.046+05:30 INFO 14688 ... [loan] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8081 (http)
2026-07-09T16:32:33.799+05:30 INFO 14688 ... [loan] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-09T16:32:33.799+05:30 INFO 14688 ... [loan] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-09T16:32:33.766+05:30 INFO 14688 ... [loan] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1128 m
s
2026-07-09T16:32:34.198+05:30 INFO 14688 ... [loan] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8081 (http) with context path '/'
2026-07-09T16:32:34.287+05:30 INFO 14688 ... [loan] [ restartedMain] com.cognizant.loan.LoanApplication : Started LoanApplication in 2.373 seconds (process running for
3.179)
2026-07-09T16:33:01.325+05:30 INFO 14688 ... [loan] [nio-8081-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-09T16:33:01.325+05:30 INFO 14688 ... [loan] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-09T16:33:01.325+05:30 INFO 14688 ... [loan] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms

PORTSDEBUG CONSOLEPROBLEMSOUTPUTTERMINAL

Spring Boot (v4.1.0)

2026-07-09T16:32:32.529+05:30 INFO 14688 ... [loan] [ restartedMain] com.cognizant.loan.LoanApplication : Starting LoanApplication using Java 23.0.1 with PID 14688 (G:\Coding\Stuff\CTIS\Stuff\Exercises\Week4\Microservices\Loan)
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3.179)
2026-07-09T16:33:01.325+05:30 INFO 14688 ... [loan] [nio-8081-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
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2026-07-09T16:33:01.325+05:30 INFO 14688 ... [loan] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms

LOAN

srcmain

java\com\cognizant\loan

controllerLoanController.java

modelLoan.javaLoanApplication.java

resources

static

templates

application.properties

test

target

.gitattributes

.gitignore

HELP.md

mvnw

mvnw.cmd

pom.xml

OUTLINE

TIMELINE

LOGICAL STRUCTURE

Ln 14, Col 1TabSize: 4UTF-8LFJava